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TECHNIZER INDIA · TRAINING TRACK

CLAUDE CODE

Complete Guide

TECHNIZER
Inspired by Impossible

VERSION
CLAUDE CODE 2026

PREPARED BY
TECHNIZER INDIA

What We'll Cover

01
What Is Claude Code?
Agentic harness · CLI architecture · model routing

02
Models & Routing Flows
Sonnet · Opus · Haiku · Ollama local inference

03
Session Management
Named sessions · resume · fork · export · context

04
Slash Commands
Built-in commands · agentic workflows · shortcuts

05
Project Setup & Skills
CLAUDE.md · settings.json · custom skills

	Claude AI	Claude Code	Claude Cowork	Claude Design
For	Everyone	Developers	Non-developers	Designers / PMs
Where	· Browser · Mobile · Desktop (Chat tab)	· Terminal CLI · Desktop (Code tab) · IDE extensions	Desktop (Cowork tab)	Browser (claude.ai/design)
Does	Chat & assist	Writes & runs code	Automates tasks & files	Creates visuals
Input	Conversation	Prompts + code	Tasks & workflows	Prompts + uploads
Output	Text / analysis	Working code	Automated actions	Prototypes / decks
Access	All plans	Pro, Max, Team, Enterprise	Beta	Research preview

Product	Status	Access
Claude AI (chat)	✅ GA	Free + all paid plans
Claude Code (CLI + desktop)	✅ GA	Pro, Max, Team, Enterprise
Claude Cowork	✅ GA (with beta features inside)	Pro, Max, Team, Enterprise
Claude Design	🔬 Research Preview	Pro, Max, Team, Enterprise — rolling out gradually
Claude for Excel / PowerPoint / Word	✅ GA	All paid plans
Claude for Outlook	🧪 Public Beta	All paid plans
Claude Security	🧪 Public Beta	Enterprise only
Claude Mythos Preview	🔒 Gated Research Preview	Invitation-only - Project Glasswing partners

What Is Claude Code?

An agentic harness — not just a model wrapper

🔑 **Key Insight**

Claude Code is a CLI tool (TypeScript / Node) that wraps an agentic loop around the Anthropic /v1/messages API.

The model handles reasoning — Claude Code handles everything else.

Agentic Loop
QueryEngine — repeats until task complete

Tool System
50+ tools: Read, Write, Bash, Git…

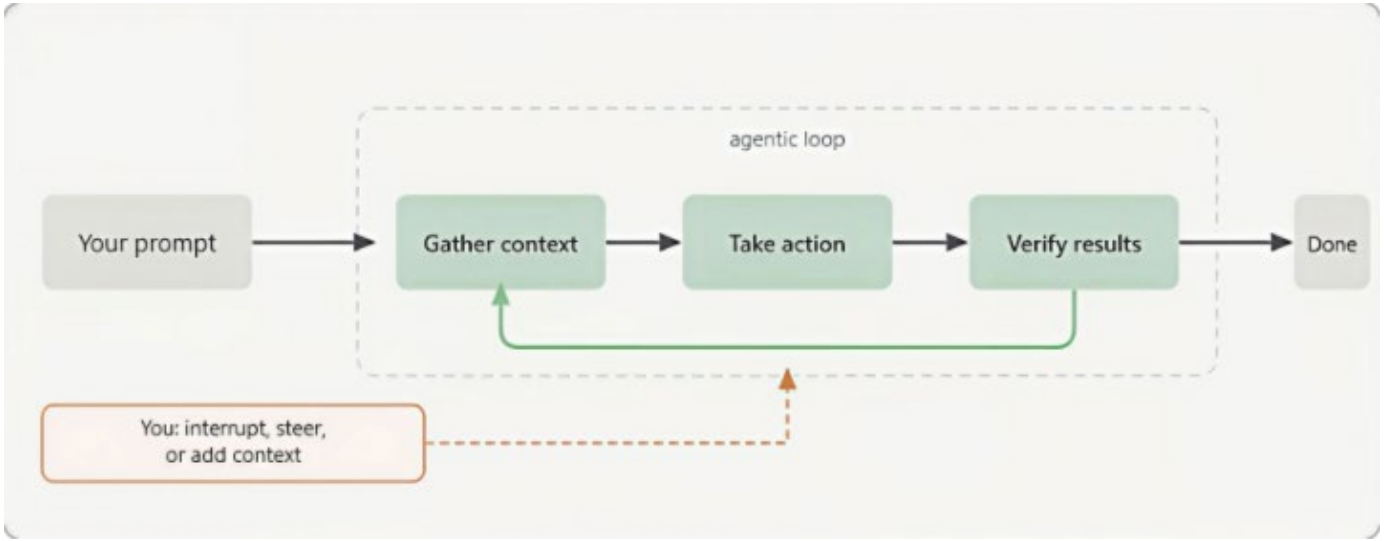
Permission Pipeline
Allow / Ask / Deny per tool

Context Manager
Compaction, pruning, context tracking

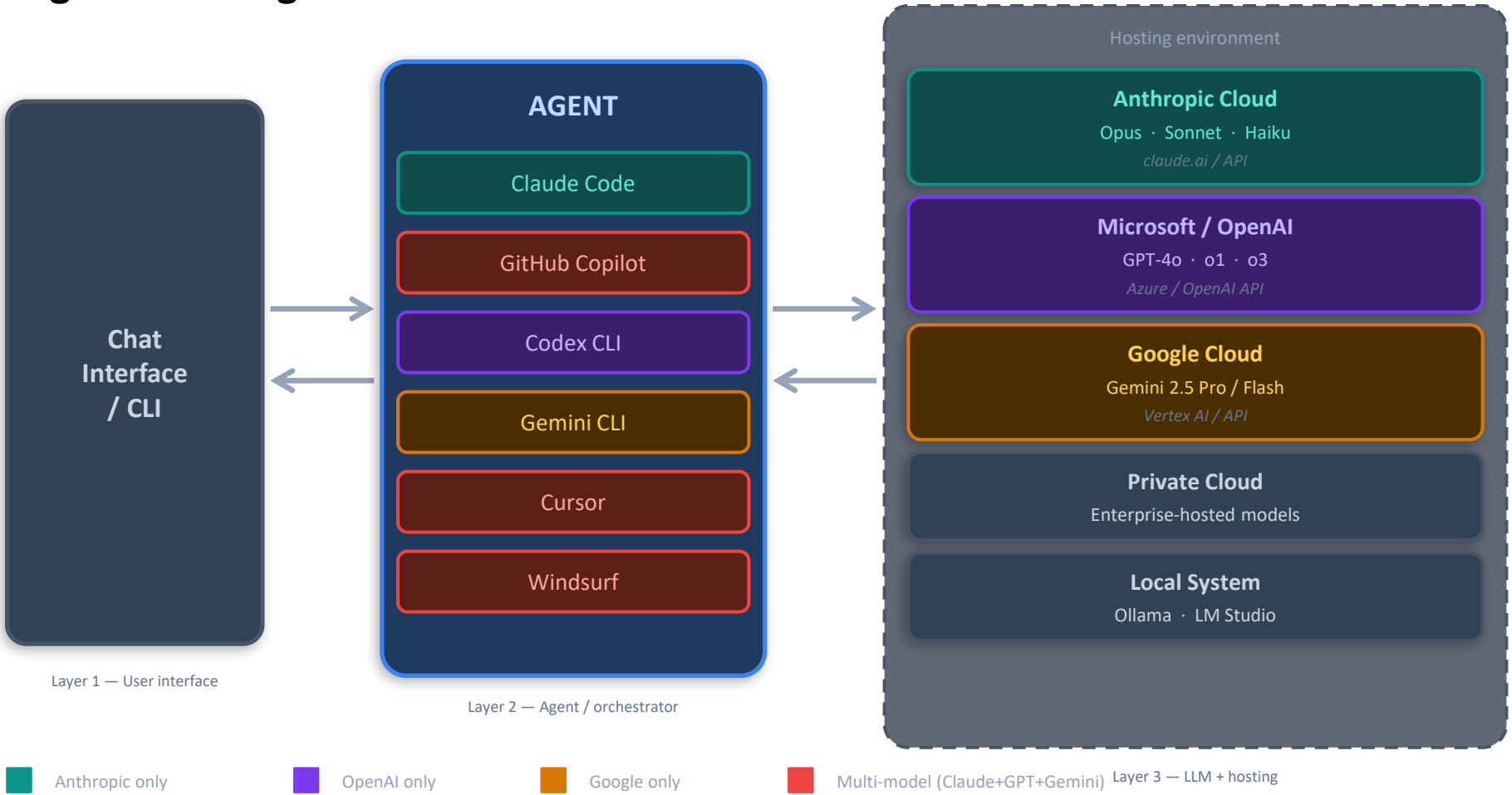
Session Persistence
Auto-saves all sessions as .jsonl

The agentic loop

The agentic loop is powered by two components: **models** that reason and **tools** that act. Claude Code serves as the **agentic harness** around Claude: it provides the tools, context management, and execution environment that turn a language model into a capable coding agent.

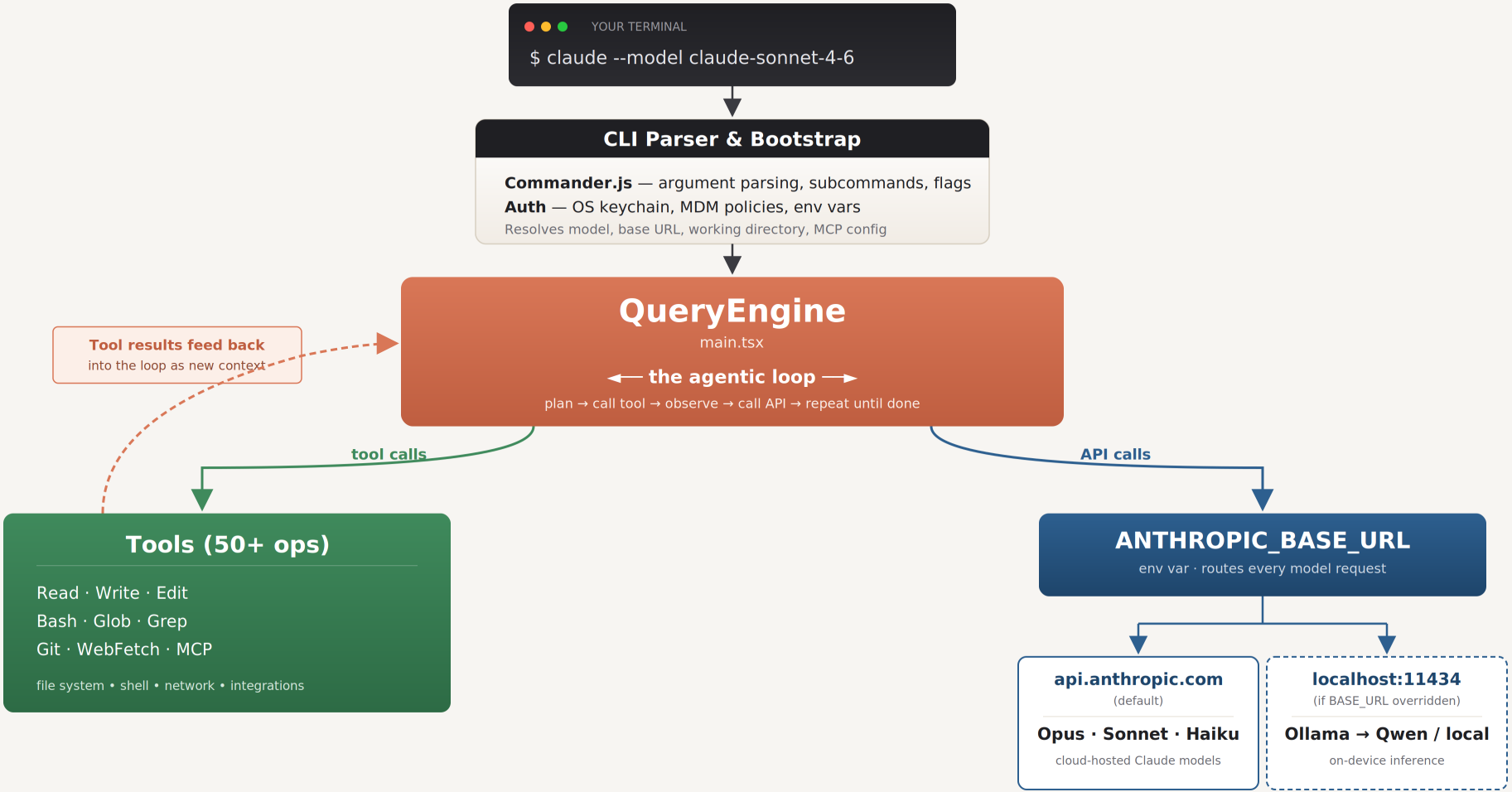


Agentic Coding Tools — Architecture



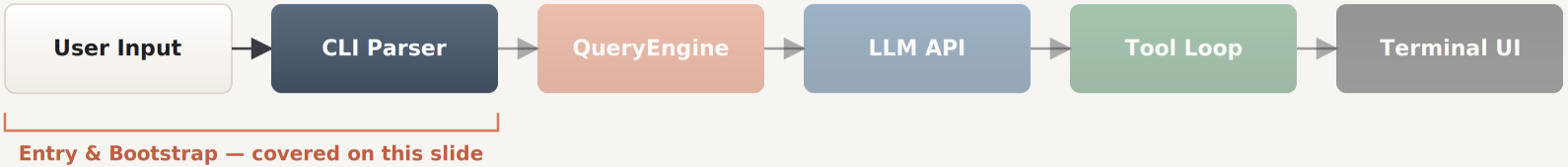
Claude Code — Runtime Architecture

From terminal command to model response, with tools and the agentic loop



Inside Claude Code — Pipeline & Bootstrap

From CLI command to ready-to-run agent



01

Entry Points

Routes to execution mode

- **CLI**
- **MCP server**
- **SDK**
- **Daemon**

Each mode has its own entry handler

02

Bootstrap

Loads runtime essentials

- **Configuration**
- **Telemetry**
- **Auth credentials**
- **Networking**

Layered: env vars · files · MDM

03

Setup

Prepares the environment

- **Working directory**
- **Hooks**
- **Plugins**
- **File watchers**

Ready before the first prompt

04

Parallel Prefetch

Fired before heavy imports

- **MDM settings**
- **Keychain**
- **API preconnect**

Cuts perceived startup latency

Parallel side-effects fire on launch — BEFORE heavy module imports —
so the CLI feels instant instead of waiting on the JS bundle.

Sources: github.com/nirholas/claude-code · zainhas.github.io · code.claude.com/docs

QueryEngine & the Agentic Loop

The heart of execution — and why it's deliberately simple

QueryEngine — the heart

Sits between the UI and the Claude API

send → respond → execute tools → return results → repeat

Three phases — they blend, and Claude course-corrects from each step

1. Gather Context

Read files, query env, explore the workspace

"What do I need to know?"

2. Take Action

Run tools, edit code, execute commands

"Make the change."

3. Verify Results

Check output, validate state, confirm intent

"Did it actually work?"

No classifiers · No RAG · No DAG

just while(tool_call) — the model itself decides every step

Sources: deepwiki.com/FlorianBruniaux · code.claude.com/docs · zainhas.github.io

The Two Routing Flows

Flow 1 — Anthropic Cloud

1. Terminal prompt
2. Claude Code CLI
3. POST api.anthropic.com/v1/messages
4. Opus / Sonnet / Haiku (cloud)
5. Stream response + tool_use
6. Execute tools locally
7. Repeat until done

Flow 2 — Ollama Local

1. Terminal prompt
2. Claude Code CLI
3. POST localhost:11434/v1/messages
4. Ollama (translates Anthropic format)
5. Local inference — GPU/CPU
6. Execute tools locally
7. Repeat until done

Cloud vs Local — Side by Side

	Flow 1 — Anthropic Cloud	Flow 2 — Ollama Local
Model Quality	Best (Opus/Sonnet/Haiku)	Good (depends on local model)
Cost	API billed per token	Free — electricity only
Privacy	Code sent to Anthropic	Fully local — never leaves machine
Speed	Fast (cloud inference)	Slower — depends on hardware
Tool Calling	Full agentic capability	Depends on local model support
Internet	Required	No — after initial model pull
Setup	Just an API key	Ollama + model download (≥16 GB RAM)

Models at a Glance

Sonnet 4.6

Daily Driver — Default

- Writing new features & components
- Fixing standard bugs & refactoring
- Tests, docs, PR reviews
- Computer use & vision tasks

Opus 4.7

Deep Thinker — Complex Work

- Hard intermittent / race condition bugs
- Architecture & system design
- Security audits & vulnerability analysis
- 1M context — entire repositories

Haiku 4.5

The Sprinter — Fast & Cheap

- Bulk find-and-replace operations
- Simple lookups & quick answers
- Boilerplate scaffolding & linting
- Rate-limit conservation tasks

opusplan

Intelligent Hybrid Mode

- Plan with Opus — execute with Sonnet
- Large feature implementations
- Complex multi-file migrations
- Best of quality + speed

Switching Models & Effort Levels

How to Switch

Mid-session

```
/model sonnet  |  /model opus  |  /model haiku
```

At startup

```
claude --model opus
```

Env var

```
export ANTHROPIC_MODEL="sonnet"
```

Settings file

```
{ "model": "sonnet" }  →  .claude/settings.json
```

One-shot

```
claude --model haiku -p "Update headers"
```

Effort Levels

`/effort low`	Fast, less thorough	Yes
`/effort medium`	Default for Pro/Max	Yes
`/effort high`	Deep reasoning	Yes
`/effort max`	Maximum — Opus 4.6 only	Session only
`/effort auto`	Reset to model default	—

💡 ultrathink — add anywhere in a prompt to trigger high effort for that single turn (Claude Code terminal only)

When to Switch — Decision Guide

Stay on Sonnet when:

- ✓ Writing features, fixing standard bugs
- ✓ Refactoring, tests, documentation
- ✓ PR creation and code reviews
- ✓ 90%+ of everyday coding tasks

Stay on Opus when:

- ✓ Hard/intermittent bugs Sonnet can't solve
- ✓ Architecture decisions & design patterns
- ✓ Security audits, unfamiliar codebases
- ✓ Tried Sonnet twice — still off-track

Stay on Haiku when:

- ✓ Bulk boilerplate / copyright header updates
- ✓ Simple find-and-replace across files
- ✓ Quick factual lookups
- ✓ Rate limit conservation

Stay on 🔄 opusplan when:

- ✓ Large features needing a thorough plan
- ✓ Complex multi-file migrations
- ✓ Want Opus intelligence + Sonnet speed
- ✓ Architecture first, code second

Session Management

All sessions auto-saved locally to ~/.claude/projects/<encoded-path>/*.jsonl — never auto-deleted. Everything restored on resume: message history, tool results, model, working directory.

Start & Name

`claude -n "feature-auth"`
Start named session

`/rename payment-v2`
Rename current session

`/rename`
Auto-name from history

`claude -n "bugfix-webhook"`
Naming convention tip

Resume

`claude -c`
Continue last session

`claude -r`
Open session picker

`claude -r "feature-auth"`
Resume by name

`/resume`
Switch mid-session

Context & Export

`/context`
Check context usage

`/compact`
Summarise older msgs

`/export payments.md`
Export as plain text

`/branch or /fork`
Fork current session

Key Shortcuts & Keyboard Controls

Keyboard Shortcuts

`Shift + Tab`
Cycle permission modes: Normal → Auto-Accept → Plan Mode

`Ctrl + O`
Toggle verbose — shows extended thinking as gray italic text

`Ctrl + G`
Open current plan in your default text editor

`Ctrl + T`
Toggle background task list display

`Option/Alt + T`
Enable / disable extended thinking

`Esc Esc`
Time machine — browse all prompts from current session

`↑ arrow`
Navigate back through past prompts (even from previous sessions)

CLI Flags at Startup

`--model <alias>`
Set model for this session

`--name / -n`
Start with a named session

`--continue / -c`
Resume last session

`--resume / -r`
Open session picker or resume by name

`--from-pr 142`
Resume session linked to a PR

`--permission-mode plan`
Start in Plan Mode

`-p "prompt"`
Headless / non-interactive mode

Project Setup — CLAUDE.md & Settings

CLAUDE.md

- Persistent memory for Claude — loaded on every session
- Define tech stack, architecture, coding standards
- Set rules: What Claude must never do
- Specify PR and Git standards
- Created with /init command
- Supports global (~/.claude/CLAUDE.md) and per-project versions

settings.json

```
{
  "model": "sonnet",
  "env": {
    "ANTHROPIC_BASE_URL":
      "http://localhost:11434",
    "ANTHROPIC_AUTH_TOKEN": "ollama",
    "ANTHROPIC_API_KEY": ""
  }
}
```

Folder Structure

```
.claude/
  settings.json       ← model, env vars
  commands/           ← custom slash cmds
  skills/             ← reusable skill files
CLAUDE.md             ← project memory
```

The 6 Building Blocks

CLAUDE.md
Project Memory

What Claude knows about your project — always. Loaded every session automatically.

Slash Commands
On-Demand Shortcuts

Reusable prompts you trigger manually with a forward slash. Personal or team-wide.

Skills
Auto-Applied Expertise

Claude reads context and invokes the right skill automatically — no prompt needed.

Agents
Specialist Instances

Isolated Claude instances with their own persona, tools, and context window.

Hooks
Automated Quality Gates

Shell commands that fire automatically around Claude actions — linting, formatting, notifications.

Plugins
Portable Team Distribution

Package all 5 layers above into a single installable unit. One command, full team setup.

Built-in Slash Commands

Session & Context

`/clear` `/compact` `/context`

`/rewind` `/rename` `/branch`

`/export` `/copy` `/resume`

Model & Config

`/model` `/effort` `/fast`

`/config` `/theme` `/output-style`

`/status` `/cost` `/usage`

Agentic Workflows

`/batch` `/simplify` `/debug`

`/loop` `/claude-api`

`/plan` `/ultraplan`

Git & Code Review

`/diff` `/security-review`

`/autofix-pr` `/install-github-app`

`/tasks` `/bashes`

Permissions & Tools

`/permissions` `/hooks` `/mcp`

`/add-dir` `/sandbox`

`/memory` `/skills` `/agents`

Utility & Info

`/help` `/doctor` `/stats`

`/insights` `/release-notes`

`/feedback` `/powerup`

Key Takeaways

1. Claude Code is an agentic harness — it executes the loop, tools, and permissions. The model (cloud or local) only handles reasoning.
2. Start with Sonnet. Escalate to Opus only when you've genuinely hit its ceiling. Use Haiku for mechanical, repetitive tasks.
3. Use opusplan for large features: plan with Opus quality, execute with Sonnet speed.
4. Name sessions early with -n. One session per task. Use /compact before context fills up. Export before big refactors.
5. Slash commands are your control panel: /model, /effort, /batch, /compact, /export, /diff — learn these first.
6. CLAUDE.md + .claude/settings.json are your project memory. Set them up once, gain consistent AI behaviour across all sessions.

Source: code.claude.com — Official Anthropic Documentation

The RTCTF Prompt Framework

Five components — for prompts Claude Code can act on

R — Role

Who Claude should be

"You are a senior TypeScript engineer..."

Sets expertise & persona

T — Task

What to actually do

"Refactor the auth module to JWT helper..."

Action verb + clear deliverable

C — Context

Background it needs

"Next.js + Prisma; tests in /tests/auth/..."

Codebase, files, patterns

T — Tone

How it should sound

"Direct; skip prose between edits..."

Voice, register, posture

F — Format

Shape of the output

"Plan first, then edit; end with diff summary"

Plan, edits, summary

A real RTCTF prompt — refactoring an auth module in Claude Code

- **R** You are a senior TypeScript engineer working in this Next.js + Prisma codebase.
- **T** Refactor the auth module to use the new JWT helper in lib/auth/jwt.ts.
- **C** All callers live in /app/api/auth/* and /pages/login. Follow patterns from lib/auth/jwt.ts.
- **T** Be direct — skip prose explanations between edits.
- **F** Show a plan first, then edit files. End with diff summary and tests to run.

Five components, five fewer guesses.

In Claude Code's tool-calling loop, every guess compounds into a real action.

RTCTF as a Design Discipline

Authoring config means writing every prompt in advance — for everyone, forever

SURFACE	RTCTF DEMANDED	PLUS
CLAUDE.md *Project's persistent identity* Miss any of the durable four — every per-turn prompt is forced to compensate.	R T C T F	— Task comes per-turn —
Slash Command *.claude/commands/refactor.md* Full RTCTF — the Task is parameterized; everything else is locked in by you.	R T C T F	+ ARGUMENTS
Skill *skills/<name>/SKILL.md* Frontmatter description is the load-trigger — bad description = wrong loads, or no loads.	R T C T F	+ TRIGGER METADATA
Subagent *.claude/agents/<role>.md* Self-contained RTCTF + which tools the agent may call (Format by exclusion).	R T C T F	+ TOOL SCOPE

Authoring is prompt engineering with permanence.

What you bake into config fires on every invocation — make sure it's the right thing.

Agents — Specialist Claude Instances

A Skill tells the main Claude how to handle a task. An Agent is a separate isolated Claude you delegate work to — its own persona, tools, and context window.

`.claude/agents/db-specialist.md`

Allowed: Read · Bash(npx prisma *) · Bash(psql *)

Senior PostgreSQL & Prisma ORM engineer. Thinks exclusively about data — schema, query performance, index strategy, migrations. Does not write application logic.

Focus Areas

- Review Prisma schema for missing indexes
- Flag N+1 patterns in Prisma calls
- Write & validate migration files
- Suggest query rewrites for performance

`.claude/agents/security-analyst.md`

Allowed: Read · Bash(npm audit) · Bash(grep *)

Senior application security engineer. Reads code exclusively through a security lens. Finds vulnerabilities and explains how to fix them. Tags every finding with OWASP category.

Focus Areas

- Authentication bypass & broken access control
- Injection vulnerabilities (SQL, command, path)
- Sensitive data exposure in logs & responses
- Insecure token handling & missing validation

Invoke with: "Ask the db-specialist to…" | "Use the security-analyst agent to audit…" | /agents (picker)

Agent vs Agentic

Two distinct AI execution patterns — know when to use each

Agent

Framework-driven orchestration

- **Orchestrator:** External framework (LangChain / LangGraph) manages the loop
- **Model-agnostic:** Swap any LLM — GPT-4, Claude, Gemini
- **Tool calling:** Framework routes tools, memory & retrieval
- **Stateful graph:** Nodes + edges define the workflow explicitly
- **Best for:** Multi-model pipelines, reusable agent graphs

e.g. LangChain · LangGraph · CrewAI · AutoGen

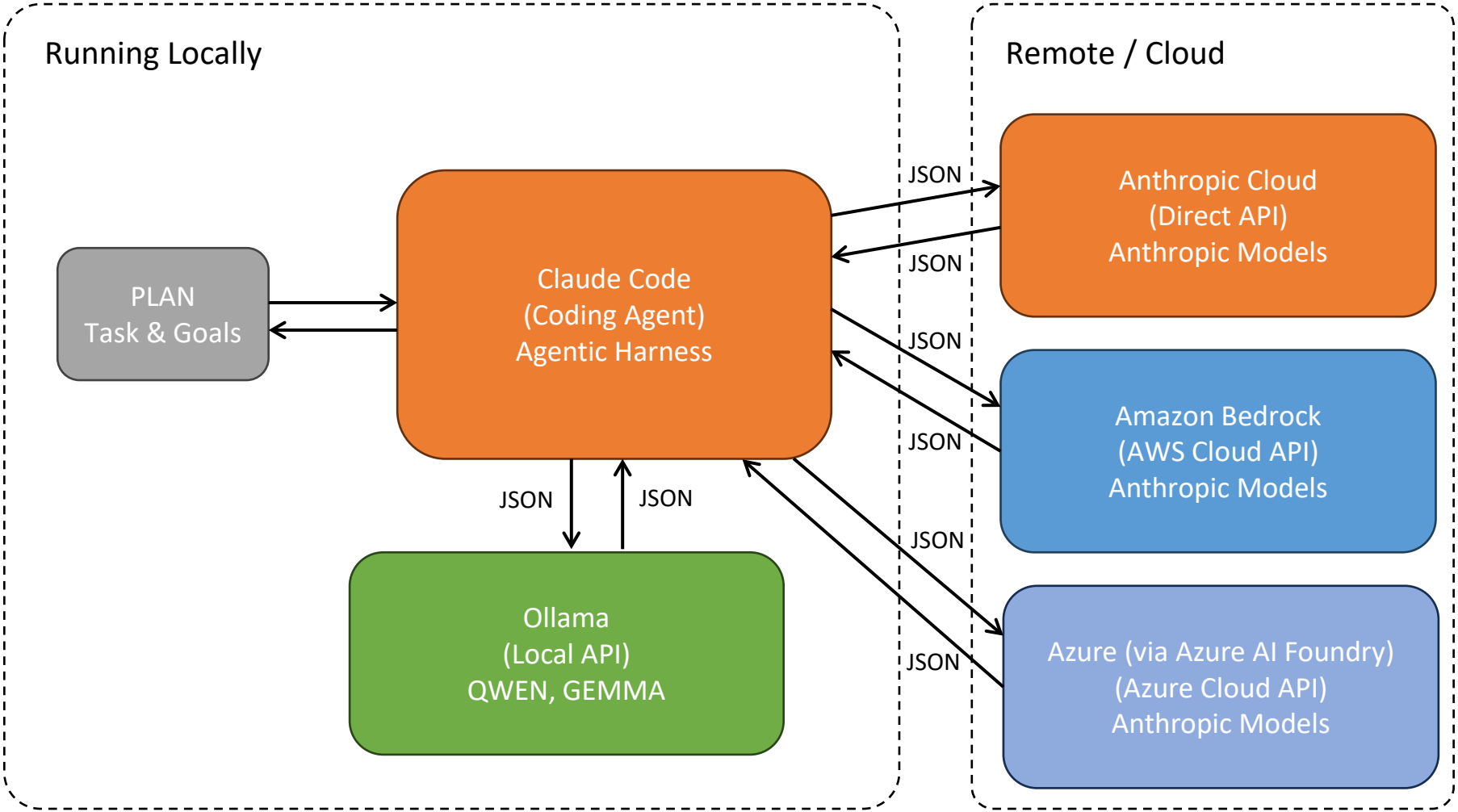
VS

Agentic

Model-native autonomous execution

- **Self-directed:** The model itself plans, decides & executes the loop
- **Model-coupled:** Claude Code / Devin drive the harness natively
- **Tool use built-in:** File system, shell, browser, APIs — all native
- **Dynamic flow:** No predefined graph; model reasons what to do next
- **Best for:** Autonomous coding, SRE tasks, open-ended workflows

e.g. Claude Code · Devin · GitHub Copilot Agent


Running Locally
Remote / Cloud
PLAN
Task & Goals
Claude Code
(Coding Agent)
Agentic Harness
Ollama
(Local API)
QWEN, GEMMA
Anthropic Cloud
(Direct API)
Anthropic Models
Amazon Bedrock
(AWS Cloud API)
Anthropic Models
Azure (via Azure AI Foundry)
(Azure Cloud API)
Anthropic Models
JSON
JSON
JSON
JSON
JSON
JSON
JSON
JSON

Agentic is a Process

The autonomous loop that runs inside every agentic system

1

👁

Perceive

Read context, files & errors

2

🧠

Plan

Decide what to do next

3

⚡

Act

Run tool, write code, call API

4

🔍

Observe

Check result & output

5

🔁

Repeat

Loop until goal achieved

↺ Autonomous loop continues until task is complete

🤖 **Agent:** The framework runs this loop externally, routing between tools and models.

⚡ **Agentic:** The model itself runs this loop natively — no framework needed.

Bot → Agent → Agentic

Each stage adds one critical capability — LLM reasoning makes it Agentic

1

Bot / Script

Rule-based automation

- Fixed if-then logic
- No reasoning ability
- Deterministic output
- RPA, decision trees

e.g. Chatbot, RPA, Macro

+ LLM as a tool

2

Agent

Framework orchestration

- Structured tool calling
- Framework drives the loop
- LLM is one of many tools
- LangChain, LangGraph

e.g. LangChain, CrewAI

+ LLM as the brain

3

Agentic

LLM IS the orchestrator

- LLM reasons & decides natively
- No external framework needed
- Perceive→Plan→Act loop
- Claude Code, Devin, Copilot

e.g. Claude Code, Devin

"When you add the capability to run an Agent based on LLM reasoning, it becomes Agentic — the LLM IS the brain."

Remember This

The three rules that separate Agent from Agentic

💡 **Rule 1: "Agent is a noun. Agentic is a verb in disguise."**

Agent

You BUILD an agent.
It is a software entity — a bot, a system, a framework.

Agentic

You RUN agentically.
It describes HOW a system operates autonomously.

💡 **Rule 2: "An Agent is what you build. Agentic is how it behaves."**

Construction

LangChain, LangGraph, CrewAI —
tools that help you assemble agents.

Behaviour

Perceive → Plan → Act → Observe → Repeat
The autonomous process loop.

💡 **Rule 3: "Agentic = a process. The model IS the orchestrator."**

Agent pattern

Framework sits between model & tools.
Orchestration is external & explicit.

Agentic pattern

Model drives the loop natively.
No middleware — Claude Code, Devin.

Agentic Workflow Commands

Bundled Skills — Claude Code v2.1.x

`/batch <instruction>`

Orchestrate large-scale changes across a codebase in parallel. Decomposes into 5–30 independent units, spawns one background agent per unit in isolated git worktrees, each opens a PR.

→ `/batch migrate src/ from Solid to React`

`/simplify [focus]`

Spawns three parallel review agents to check recently changed files for reuse, quality, and efficiency — then applies the fixes automatically.

→ `/simplify focus on memory efficiency`

`/loop [interval] [prompt]`

Run a prompt repeatedly while the session stays open. Claude self-paces if no interval is given. Great for polling or watching long-running processes.

→ `/loop 5m check if the deploy finished`

`/autofix-pr [prompt]`

Spawns a web session watching the current PR for CI failures and reviewer comments, then pushes fixes automatically. Requires the gh CLI.

→ `/autofix-pr fix lint errors in the auth module`

`/ultraplan <prompt>`

Draft a plan in an ultraplan session, review it in the browser, then execute remotely or send back to terminal. Combines deep planning with remote execution.

→ `/ultraplan redesign the payment service`

Commands vs Skills vs Agents

Decision framework — which to use when

Slash Command

`Trigger: You type /command`

Use when you want a repeatable saved prompt triggered manually

Scope: Per-project (.claude/commands/) or Personal (~/.claude/commands/)

Examples

→ /commit /security-review /fix-issue /standup

Skill

`Trigger: Claude reads context and invokes automatically`

Use when you want Claude to apply expertise based on what it sees — without being asked

Scope: Per-project (.claude/skills/) or via Plugin

Examples

→ code-review skill fires when you say 'check the code'

→ write-tests skill fires when new code is produced

Agent

`Trigger: You delegate: 'use the X agent to…'`

Use when you want full isolation — its own persona, tool set, context — for a focused specialist task

Scope: Per-project (.claude/agents/) or via Plugin

Examples

→ db-specialist reviews the Prisma schema

→ security-analyst audits authService.js

Claude Code

Hooks Reference

Complete event catalog with matcher values · v2.1.x

Hook Event Catalog — by Category

Session	SessionStart · Setup · SessionEnd
Turn	UserPromptSubmit · UserPromptExpansion · Stop · StopFailure
Tools	PreToolUse · PostToolUse · PostToolUseFailure · PostToolBatch · PermissionRequest · PermissionDenied
Agent / Task	SubagentStart · SubagentStop · TeammateIdle · TaskCreated · TaskCompleted
Context / Config	InstructionsLoaded · ConfigChange · CwdChanged · FileChanged
Compaction	PreCompact · PostCompact
Worktree	WorktreeCreate · WorktreeRemove
MCP / Elicitation	Elicitation · ElicitationResult
Display	MessageDisplay · Notification

30 events total

How a hook event flows

from Claude Code firing it, through to your script reading the data and responding.

WHAT TRIGGERS A HOOK

Claude Code reaches a lifecycle boundary (e.g. a tool is about to run). It pauses, spawns your script as a subprocess, and feeds a `JSON` blob to its `stdin`. Your script has full control over what happens next.

WHAT YOUR SCRIPT RETURNS

Two channels back: **exit code** (0 = continue, 2 = block/refuse) and optional **stdout JSON** with a structured decision, reason, and context for Claude to read.

Matcher Values — What Each Hook Filters By

Hook event(s)	Matcher filters by
PreToolUse, PostToolUse, PostToolUseFailure, PermissionDenied	*Tool name e.g. Bash | Write|Edit | Write|Edit|MultiEdit*
SessionStart	*startup | resume | clear | compact*
SessionEnd	*clear | resume | logout | prompt_input_exit*
Setup	*init | maintenance*
Notification	*permission_prompt | idle_prompt | auth_success | elicitation_dialog | elicitation_complete | elicitation_response.*
SubagentStart / SubagentStop	*Agent type*
PreCompact / PostCompact	*manual | auto*
ConfigChange	*Config source*
FileChanged	*Filename pattern (glob/regex)*
StopFailure	*Error type*
InstructionsLoaded	*Load reason*
UserPromptExpansion	*Command name*
Elicitation / ElicitationResult	*MCP server name*

Hooks With No Matcher Support

These 10 hooks always fire — they cannot be scoped to a subset of events

UserPromptSubmit
Fires before every user prompt — inject context or block

PostToolBatch
After a parallel tool batch resolves — exit 2 stops the loop

Stop
End of turn — exit 2 forces Claude to keep working

TeammateIdle
Agent team member idle notification

TaskCreated
Model creates a task

TaskCompleted
A task finishes

WorktreeCreate
Git worktree created

WorktreeRemove
Git worktree removed

CwdChanged
Working directory changed

MessageDisplay
Assistant text about to be shown — can rewrite output

For these hooks, filter inside your script by reading tool_name or other fields from stdin JSON — not via the matcher field.

What are Claude Code Hooks?

User-defined handlers that fire automatically at lifecycle events — defined in .claude/settings.json

command

Shell script — reads event JSON from stdin, signals via exit code and stdout JSON. Covers ~90% of use cases.

http

Webhook POST — sends the event JSON as the request body to an HTTP endpoint.

prompt

LLM evaluates a yes/no question. Best for Stop and SubagentStop validation.

agent

Spawns a full subagent with tool access for deep analysis or validation.

mcp_tool

Invokes a named tool on a connected MCP server directly. Requires server + tool fields.

A system prompt is a request. A hook is a guarantee.

settings.json Structure

Matcher is always at the outer object level — never inside the hooks array

```
"hooks": {
  "PostToolUse": [
    {
      "matcher": "Write|Edit",        // outer level
      "hooks": [
        {
          "type": "command",          // command | http | prompt | agent
          "command": "node .claude/hooks/eslint-fix.js",
          "timeout": 30
        }
      ]
    }
  ]
}
```

`.claude/settings.json`
Project-level — committed to git, shared with team

`.claude/settings.local.json`
Local overrides — gitignored, personal hooks only

`~/.claude/settings.json`
User global — applies to all projects

Hook Quick Reference

Use case	Hook	Matcher	Notes
Auto-format on file write	**PostToolUse**	*Write|Edit*	Run prettier/eslint — fires after tool succeeds
Block dangerous commands	**PreToolUse**	*Bash*	Exit 2 to block — reads stdin JSON for command text
Inject context at session start	**SessionStart**	*startup*	stdout fed into Claude's context automatically
Force tests before turn ends	**Stop**	—	Exit 2 to keep Claude working; exit 0 to allow stop
Guard context before compaction	**PreCompact**	—	Inject must-survive context before transcript shrinks
Block prompt to frozen module	**UserPromptSubmit**	—	Read prompt from stdin; exit 2 to block entirely

Hooks — Automated Quality Gates

Shell commands that fire automatically around Claude actions — defined in .claude/settings.json

PostToolUse

Fires AFTER Claude writes, edits, or runs something.
Most common use: auto-lint or format every file Claude touches.

PreToolUse

Fires BEFORE Claude runs a command. Can block the action.
Use for approval gates on sensitive operations.

Stop

Fires when Claude finishes a task.
Use for desktop notifications so you can step away during long tasks.

Notification

Fires when Claude needs your attention mid-task.
Route to Slack, email, or desktop notification system.

```
"hooks": {
  "PostToolUse": [{ "matcher": "Write|Edit",   "hooks": [{ "type":"command", "command":"xargs npx eslint --fix" }] }],
  "PostToolUse": [{ "matcher": "Write(*.py)",  "hooks": [{ "type":"command", "command":"xargs python -m black" }] }],
  "Stop":        [{ "hooks": [{ "type":"command", "command": "osascript -e 'display notification "Claude finished"'" }] }]
}
```

Here are all 6 Notification matcher types:

Matcher	When it fires	Feasibility for Slack
permission_prompt	Claude needs permission for a tool call	**High** — audit trail, alerts team to review
idle_prompt	Claude is idle / waiting	**High** — great for alerting "session is hanging"
auth_success	Successful authentication	**Medium** — low volume, meaningful signal
elicitation_dialog	MCP server requests user input	Low — noisy, MCP-specific
elicitation_complete	MCP elicitation finishes	Low — noisy, rapid-fire
elicitation_response	User responds to elicitation	Low — noisy, rapid-fire

Plugins — Package & Share Your Setup

Without plugins: new developers get a Slack message — "copy this folder, edit this JSON, install these servers." Two hours later, everyone has a slightly different setup. Plugins make your entire Claude Code configuration installable in one command.

📦 Plugin Structure

```
my-team-plugin/
├── .claude-plugin/
│   └── plugin.json        ← Manifest ONLY goes here
├── commands/              ← Slash commands
├── agents/                ← Specialist agents
├── skills/                ← Skills
├── hooks/
│   └── hooks.json         ← Hooks config
└── .mcp.json              ← MCP servers
```

⚙️ plugin.json Manifest

```
{
  "name": "my-team-plugin",
  "version": "1.0.0",
  "description": "Standard Claude Code setup
                  for our engineering team",
  "commands": ["commands/"],
  "agents":   ["agents/"],
  "skills":   ["skills/"],
  "hooks":    "hooks/hooks.json"
}
```

Clone repo → **Run claude** → **/plugin install [url]** → **Type /help — done**

Four steps. Fifteen minutes. Every developer has the exact same environment.

What is MCP?

and why it changes how AI works with the world

WHAT IS MCP

Model Context Protocol

An open standard by Anthropic (Nov 2024)

> "Think of MCP as USB-C for AI —
> one universal plug for every tool."

AI model (Claude) acts as a **MCP Client**

External tools expose themselves as **MCP Servers**

A shared protocol bridges them — **no custom glue code needed**

Claude ⇔ MCP Client ⇔ MCP Server ⇔ Tool

WHY MCP

01 No more custom integrations

Before MCP, every tool needed bespoke glue code per AI app.

02 Secure by design

Servers declare capabilities; clients request with explicit consent.

03 Composable & reusable

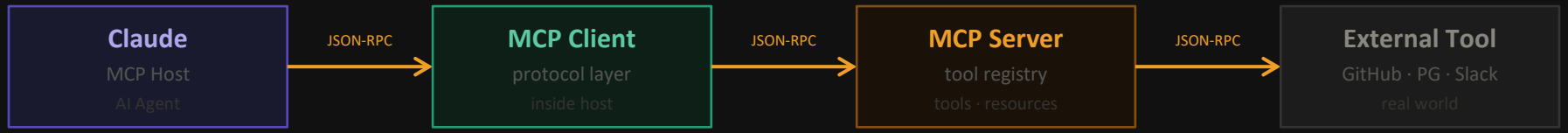
Build one MCP server, plug it into any AI host instantly.

04 Live context, live actions

AI reads real-time data and takes real actions — not just text.

How MCP Communicates

Request → Tool Call → Response — via JSON-RPC over stdio or HTTP/SSE



← response flows back as tool result

GitHub MCP

@modelcontextprotocol/server-github

TOOLS

create_issue
list_pull_requests
create_or_update_file
search_repositories

"Create a PR for my bug fix branch"
→ calls create_pull_request

PostgreSQL MCP

@modelcontextprotocol/server-postgres

TOOLS

query (read-only SQL)
list_tables
describe_table
get_schema_info

"Show top 10 users by revenue"
→ calls query → SELECT ...

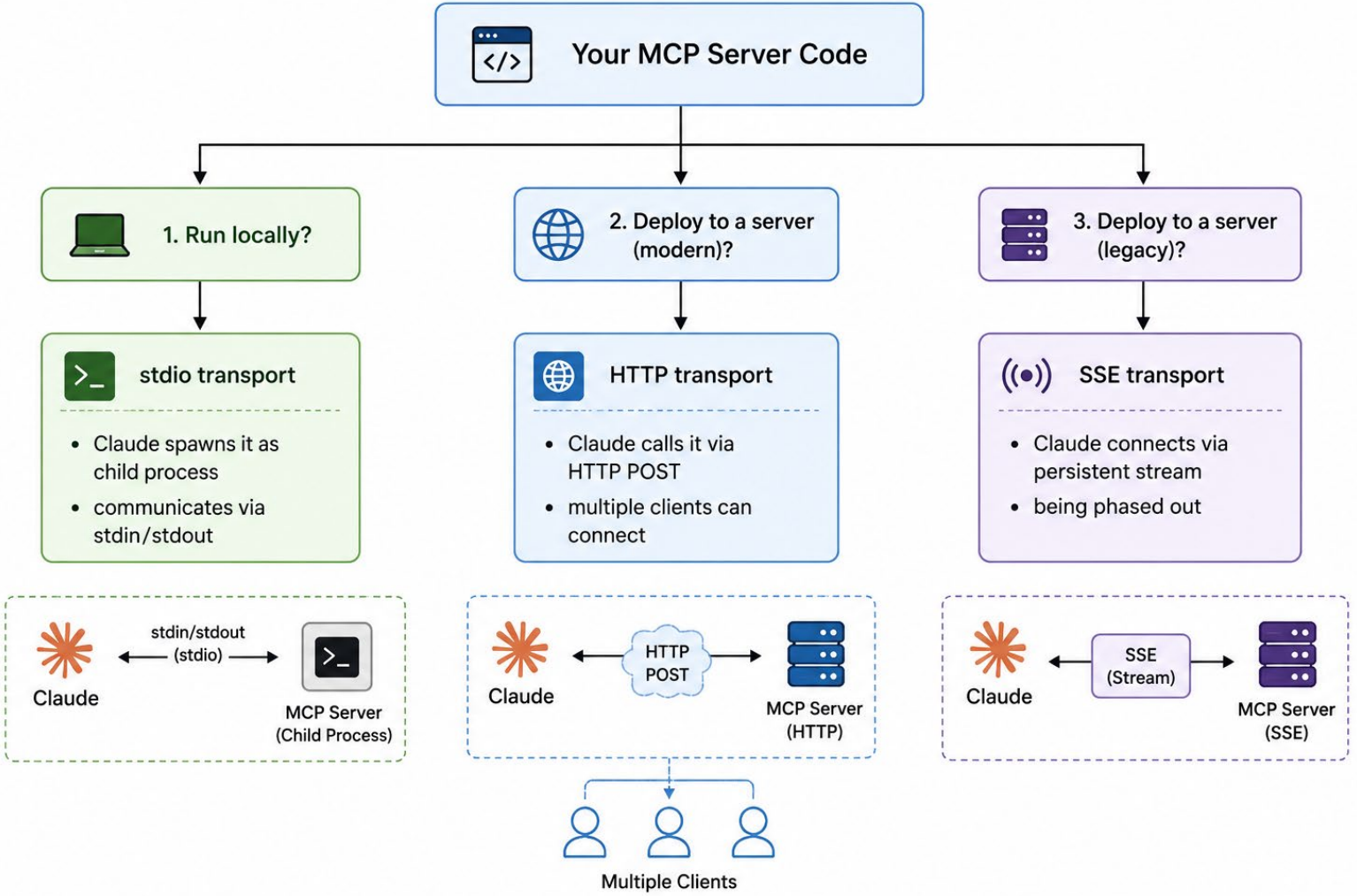
Slack MCP

@modelcontextprotocol/server-slack

TOOLS

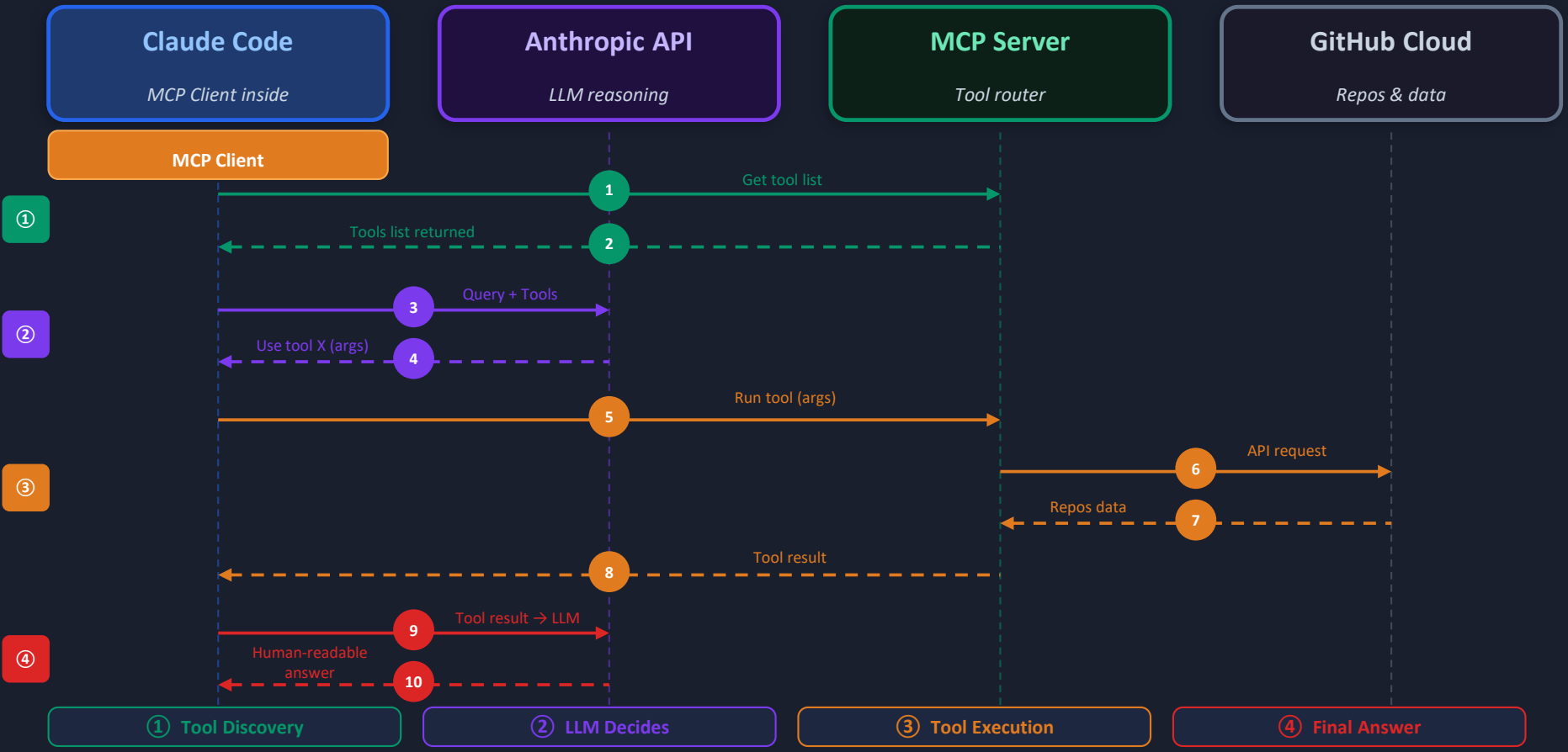
post_message
list_channels
get_channel_history
reply_to_thread

"Post standup summary to #dev"
→ calls post_message


Your MCP Server Code
1. Run locally?
2. Deploy to a server (modern)?
3. Deploy to a server (legacy)?
stdio transport
Claude spawns it as child process
communicates via stdin/stdout
HTTP transport
Claude calls it via HTTP POST
multiple clients can connect
SSE transport
Claude connects via persistent stream
being phased out
Claude
stdin/stdout (stdio)
MCP Server (Child Process)
Claude
HTTP POST
MCP Server (HTTP)
Multiple Clients
Claude
SSE (Stream)
MCP Server (SSE)

MCP Architecture — How Claude Code Talks to GitHub

"Show me all repositories in my GitHub account"



MCP Flow: 12 Steps Explained

1. ① Tool Discovery
   Claude Code needs a list of tools to send to the LLM
2. ① Tool Discovery
   MCP Client talks to MCP Server to get list of available tools
3. ② LLM Decides
   Claude Code sends the query + Tools to Anthropic API (LLM)
4. ② LLM Decides
   Anthropic API returns which tool to use + arguments needed
5. ③ Tool Execution
   Claude Code runs the tool with the returned arguments
6. ③ Tool Execution
   MCP Client sends call tool request to MCP Server
7. ③ Tool Execution
   MCP Server requests GitHub Cloud to provide repository details
8. ③ Tool Execution
   GitHub Cloud returns response data to MCP Server
9. ③ Tool Execution
   MCP Server returns tool call response back to MCP Client
10. ④ Final Answer
    Claude Code receives the result of running the tool
11. ④ Final Answer
    Tool result is sent to Anthropic API (LLM) for formatting
12. ④ Final Answer
    LLM formats and returns human-readable output to Claude Code

Full Bug Lifecycle — End to End with MCP

1	**/security-review** Claude audits authService.js	Finds: refresh token store has no expiry cleanup — stale tokens accumulate indefinitely (memory leak + security risk)
2	**/create-issue** Claude drafts & posts a structured GitHub issue via MCP	Returns a live GitHub URL. Open it in browser to confirm — full structured issue: summary, behaviour, files, acceptance criteria. No browser tab opened by you.
3	**/fix-issue 1** Claude reads issue #1 from GitHub via MCP, locates code, implements fix	Adds cleanupExpiredTokens() to authService.js. Writes a unit test for it. References the issue description to guide the fix.
4	**/commit** Claude runs git diff --staged, reads what changed, generates commit message	Produces: fix(auth): remove stale refresh tokens on expiry (closes #1) — reads before it writes.
5	**/standup** Claude reads today's git log, sees the commit, drafts standup update	Ready to paste into Slack or Teams. Yesterday / Today / Blockers — factual, 3 bullets max.

Team Rollout — Step by Step

Licensing & Seat Assignment

Assign Standard or Premium seats from the admin panel. Users authenticate with their org account — no individual API keys needed.

Standard = included limits · Premium = higher limits for power users

Create Team Configuration Repository

Dedicated Git repo for Claude Code config: plugin, shared CLAUDE.md templates, approved MCP server list. Separate from product code.

Structure: plugin/ · templates/ · README.md

Write CLAUDE.md Templates Per Project Type

Create templates for each project type (Node.js microservice, Python pipeline, React frontend…). New projects start from the template.

Commit CLAUDE.md to each product repo so every team member gets it

Build & Publish the Plugin

Package commands, skills, agents, and hooks into the plugin. Push to GitHub or internal Git host — this is your team marketplace URL.

/plugin install [url] → instant full setup for any developer

Admin Policy Setup

Push managed settings to all users via admin panel: approved MCP servers, deny list, team marketplace URL. Developers cannot override managed settings.

"managedSettings": { "deniedMcpServers": ["*personal*"], "deny": ["Read(./.env*)"] }

Developer Onboarding: 4 Steps, 15 Minutes

Install Claude Code → authenticate → clone repo → open project → /plugin install [url] → /help to verify team commands appear.

Every developer has the exact same environment from day one

Q&A Scenarios — Common Questions

Most frequent audience questions and suggested trainer responses

Q: Is our source code sent to Anthropic?

A: By default, yes — to provide responses. Enterprise: zero data retention option via API. The deny list in settings.json blocks Claude from reading secrets or proprietary files.

Q: What if Claude makes a mistake and breaks code?

A: Three safeguards: (1) Claude shows its plan before destructive operations. (2) Always work in a branch — never on main. (3) Every change is tracked via Git — git diff and git checkout bring you back.

Q: We use GitHub Copilot already. Why Claude Code?

A: Different tools. Copilot = inline completion while typing. Claude Code = terminal agent that plans, reasons across files, and executes sequences. Many teams use both — they're complementary.

Q: How much does this cost for 50 developers?

A: Team plan: Standard seats (included limits) + Premium seats (higher limits). extra-usage option with per-user spending caps. Run a pilot with a small group to understand actual consumption first.

Q: Can Claude Code access our internal documentation?

A: Yes — via MCP. If your documentation system has an MCP server, Claude queries it directly, reasoning across code and docs simultaneously. Powerful for onboarding and architecture questions.

Let’s Connect with us!

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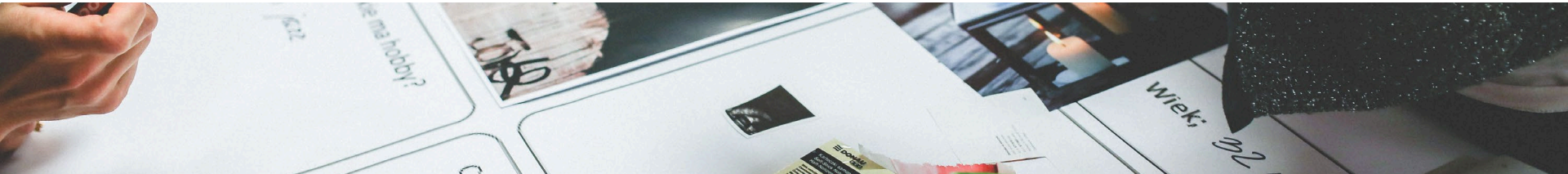
Reach out for collaborations, inquiries, or expert technology consulting.



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THANK YOU